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Dump trailer and system for a semi-trailer truck

Abstract

A dump trailer and system for a semi-trailer truck are disclosed. The dump trailer and system includes a tub, a coupling plate configured to engage the dump trailer to the tractor, a first hydraulic cylinder hinge, and a second hydraulic cylinder that cooperate to improve manufacturing economics and/or dumping of a load carried by the dump trailer.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is Continuation-in-Part (CIP) of U.S. patent application Ser. No. 17/208,032 entitled “DUMP TRAILER AND SYSTEM FOR A SEMI-TRAILER TRUCK,” and filed 22 Mar. 2021, which is a continuation of U.S. patent application Ser. No. 17/080,511 (also referred to as U.S. Pat. No. 10,988,067 issued on 27 Apr. 2021) entitled “DUMP TRAILER AND SYSTEM FOR A SEMI-TRAILER TRUCK”, and filed 26 Oct. 2020. The present application also claims priority to U.S. Provisional Application No. 63/335,194, filed on 26 Apr. 2022, entitled “DUMP TRAILER AND SYSTEM FOR A SEMI-TRAILER TRUCK.” The entirety of the aforementioned applications are incorporated herein by reference for all purposes.

TECHNICAL FIELD

(1) This disclosure pertains generally, but not by way of limitation, to a dump trailer and system for a semi-trailer truck. In particular, the dump trailer and system for a semi-trailer truck described herein provide basic operational benefits to the field of moving cargo from one location to another location with a semi-trailer truck.

BACKGROUND

(2) A semi-trailer truck is the combination of a tractor unit and a semi-trailer to carry freight. The tractor unit is coupled to the trailer with a hitch called a fifth-wheel; and the semi-trailer is configured for the freight being carried. A semi-trailer may, for example, be configured as a dump trailer having a closed end capable of being raised to allow cargo (e.g., road materials or agricultural produce) to slide out an open end that is opposite of the raised closed end. Other configurations include: a box with access doors on the back and/or sides; a car carrier for transporting vehicles; a flatbed trailer for carrying large pieces of equipment; a drop-deck trailer; a double decker configured to carry pallets of cargo; a dry bulk trailer for carrying dry powder materials such as cement, sand, and flour; a livestock trailer for hauling livestock; and, a tanker trailer are used for hauling liquids such as gasoline.

(3) In the case of a semi-trailer truck configured with a dump trailer, typically the dump trailer is loaded with cargo at one location and then transported to a different location. At this different location, the closed end is raised to allow the cargo to be ejected from the dump trailer.

SUMMARY

(4) In one example, a dump trailer for a semi-trailer truck may include: a tub, defining an exterior portion that may include: a web; a left wall adjoining the web; a right wall adjoining the web and symmetrical to the left wall; a closed end adjoining the web, the left wall, and the right wall; an open end oppositely disposed from the closed end and adjoining the web, the left wall, and the right wall; a hinge axis parallel to and offset from the web; a left hinge formed on the exterior portion of

the left wall and coaxial with the hinge axis; a right hinge formed on the exterior portion of the right wall and coaxial with the hinge axis; a distal draft arm axis parallel and adjacent to the web, the distal draft arm axis formed on the exterior portion of the web; and a center hinge formed on the web and coaxial with the distal draft arm axis, the center hinge symmetrically disposed between the left wall and the right wall; a coupling plate to engage the dump trailer to the semi-trailer truck, may include: a hydraulic cylinder axis formed on the coupling plate; a left hydraulic cylinder hinge formed on the coupling plate and coaxial with the hydraulic cylinder axis; a right hydraulic cylinder hinge formed on the coupling plate and coaxial with the hydraulic cylinder axis; a proximal draft arm axis formed on the coupling plate; and a draft arm hinge formed on the coupling plate and coaxial with the draft arm axis, the draft arm hinge symmetrically disposed between the left hydraulic cylinder hinge and the right hydraulic cylinder hinge; a draft arm symmetrically disposed between the left wall and the right wall of the tub, the draft arm may include: a proximal end attached to the draft arm hinge of the coupling plate and rotationally attached to the coupling plate about the proximal draft arm axis; and a distal end oppositely disposed from the proximal end, attached to the center hinge of the tub, and rotationally attached to the tub about the distal draft arm axis; a left hydraulic cylinder defining a left cylinder length, the left hydraulic cylinder may include: a proximal end attached to the left hydraulic cylinder hinge; a distal end oppositely disposed from the proximal end and attached to the left hinge on the tub; and wherein addition of a first amount of fluid changes the left cylinder length; a right hydraulic cylinder defining a right cylinder length, the right hydraulic cylinder may include: a proximal end attached to the right hydraulic cylinder hinge; a distal end oppositely disposed from the proximal end and attached to the right hinge on the tub; and wherein addition of a second amount of fluid changes the right cylinder length; a rear assembly pivotally attached to the tub at the web adjacent to the open end, the rear assembly may include: a first pair of wheels; and a rear pivot around which the tub rotates; whereby increase in both left cylinder length and right cylinder length cause: the distal end of the draft arm to move away from earth; and the tub to rotate about the rear pivot.

(5) In another example, a method of controlling unloading of a dump trailer may include: providing a tub, defining an exterior portion that may include: a web; a left wall adjoining the web; a right wall adjoining the web and symmetrical to the left wall; a closed end adjoining the web, the left wall, and the right wall; an open end: oppositely disposed from the closed end; and, adjoining the web, left wall, and right wall; a hinge axis formed through the left wall and the right wall; a left hinge formed on the exterior portion of the left wall and coaxial with the hinge axis; a right hinge formed on the exterior portion of the right wall and coaxial with the hinge axis; a distal draft arm axis adjacent to the web and on the exterior portion of the web; a center hinge formed on the web and coaxial with the distal draft arm axis; and a center plane symmetrically disposed between the left wall and the right wall; providing a coupling plate, configured to engage the dump trailer to semi-trailer truck, may include: a hydraulic cylinder axis formed on the coupling plate; a left hydraulic cylinder hinge: formed on the coupling plate; and coaxial with the hydraulic cylinder axis; a right hydraulic cylinder hinge: formed on the coupling plate; and coaxial with the hydraulic cylinder axis; a proximal draft arm axis formed on the coupling plate; and a draft arm hinge: formed on the coupling plate; and coaxial with the draft arm axis; providing a draft arm, symmetrically disposed between the left wall and the right wall of the tub, may include: a proximal end rotationally attached to the coupling plate about the proximal draft arm axis; and a distal end: oppositely disposed from the proximal end; attached to the center hinge; and whereby the draft arm is rotationally attached to the tub about the distal draft arm axis; providing a left hydraulic cylinder, defining a left cylinder length, may include: a proximal end attached to the left hydraulic cylinder hinge; a distal end oppositely disposed from the proximal end and attached to the left hinge on the tub; and wherein addition of a first amount of fluid changes the left cylinder length; providing a right hydraulic cylinder, defining a right cylinder length, may include: a proximal end attached to the right hydraulic cylinder hinge; a distal end oppositely disposed from the proximal end and

attached to the right hinge on the tub; and wherein addition of a second amount of fluid changes the right cylinder length; providing a rear assembly attached to the tub adjoining the web and adjacent to the open end, the rear assembly may include: a first pair of wheels; and a rear pivot about which the tub rotates; a second pair of wheels parallel to the first pair of wheels; a spring assembly may include: at least one leaf spring may include first end and a second end; a first axle attachment on the first end; a second axle attachment on the second end; and a hinge symmetrically attached between the first end and the second end of the leaf spring; wherein the first pair of wheels are attached to the first axle attachment with a first axle; wherein the second pair of wheel are attached to the second axle attachment with a second axle; and wherein the rear pivot about which the tub rotates is coaxial with the hinge of the rear assembly; increasing in both the left cylinder length and the right cylinder length: moves the distal end of the draft arm to away from earth; and rotates the tub about the rear pivot; monitoring perpendicularity of the center plane; increasing either the left cylinder length or the right cylinder length to compress the leaf spring in the spring assembly; and thereby controlling the unloading of the dump trailer.

(6) In another example, a dump trailer for a semi-trailer truck may include: a chassis plane formed on the semi-trailer truck, wherein the chassis plane is defined by at least two axles of the semi-trailer truck that are coplanar to the chassis plane; a tub, may include: a web; a left side adjoining the web; a right side adjoining the web; and a center plane formed on the tub, wherein the center plane is symmetrically located between the left wall and the right wall; a draft arm symmetrically disposed between the left side and the right side of the tub, the draft arm may include: a proximal end; and a distal end oppositely disposed from the proximal end, rotationally attached to the web of the tub; a coupling plate to engage the dump trailer to the semi-trailer truck, wherein: the proximal end of the draft arm is rotationally attached to the coupling plate; a left hydraulic cylinder defining a left cylinder length, the left hydraulic cylinder may include: a proximal end rotationally attached to the coupling plate; and a distal end oppositely disposed from the proximal end and rotationally attached to the tub; a right hydraulic cylinder defining a right cylinder length, the right hydraulic cylinder may include: a proximal end rotationally attached to the coupling plate; and a distal end oppositely disposed from the proximal end and rotationally attached to the tub; a rear assembly pivotally attached to the tub at the web, the rear assembly may include: a first pair of wheels; and a rear pivot around which the tub rotates; a perpendicular condition wherein: the center plane of the tub is perpendicular to the chassis plane; and the left cylinder length and the right cylinder length are equal; a canted condition wherein: the center plane is non-perpendicular to the chassis plane; and the right cylinder length and the right cylinder length are different.

Description

BRIEF DESCRIPTION OF THE DRAWING

(1) The accompanying figures of the drawing, which are included to provide a further understanding of general aspects of the dump trailer and system for a semi-trailer truck, are incorporated in and constitute a part of this specification. These illustrative aspects of the dump trailer and system for a semi-trailer truck, and together with the detailed description, explain the principles of the system, components, and associated methods. No attempt is made to show structural details of the invention in more detail than may be necessary for a fundamental understanding of the system and various ways in which it may be practiced. The following figures of the dump trailer and system for a semi-trailer truck include:

- (2) FIG. 1 is a semi-trailer truck having a tractor unit provided with a dump trailer;
- (3) FIG. 2 is a side elevation view of a semi-trailer truck in a transporting condition;
- (4) FIG. 3 is a side elevation view of a semi-trailer truck in a dumping condition;
- (5) FIG. 4 is a perspective view of a top-front side of a tub of a dump trailer;

- (6) FIG. 5 is a side elevation view of the left side of the tub of a dump trailer;
- (7) FIG. 6 is a front elevation view of the closed end a tub of a dump trailer;
- (8) FIG. 7 is a bottom plan view of a tub of a dump trailer;
- (9) FIG. 8 is a top perspective view of a rear assembly of a dump trailer;
- (10) FIG. 9 is a bottom perspective view of a rear assembly of a dump trailer;
- (11) FIG. 10 is a perspective view of a left spring assembly of a dump trailer;
- (12) FIG. 11 is a side view of a left spring assembly of a dump trailer;
- (13) FIG. 12 is a perspective view of a right spring assembly of a dump trailer;
- (14) FIG. 13 is a side view of a right spring assembly of a dump trailer;
- (15) FIG. 14 is a a of a side elevation view of a rear assembly dump trailer;
- (16) FIG. 15 is a perspective view of a bottom of a pyramid of a dump trailer;
- (17) FIG. 16 is a rear elevation view of the rear assembly of a dump trailer;
- (18) FIG. 17 is a perspective view of a coupling plate of a dump trailer;
- (19) FIG. 18 is a rear elevation view of the coupling plate of a dump trailer;
- (20) FIG. 19 is a front top perspective view of a draft arm assembly of a dump trailer;
- (21) FIG. 20 is a top plan view of the draft arm assembly of a dump trailer;
- (22) FIG. 21 is a perspective view of a left hydraulic cylinder of a dump trailer;
- (23) FIG. 22 is an assembled condition of the left hydraulic cylinder of a dump trailer;
- (24) FIG. 23 is a perspective view of a right hydraulic cylinder (in an exploded condition) of a dump trailer;
- (25) FIG. 24 is an assembled condition of the right hydraulic cylinder of a dump trailer;
- (26) FIG. 25 is a bottom front perspective view of a semi-trailer truck;
- (27) FIG. 26 is a side perspective view of a semi-trailer truck;
- (28) FIG. 27 is a front left side perspective view of the semi-trailer truck;
- (29) FIG. 28 is a side elevation view of the semi-trailer truck;
- (30) FIG. 29 is a side elevation view of a dump trailer;
- (31) FIG. 30 is a perspective view of a control system;
- (32) FIG. 31 is a top plan view of a control system;
- (33) FIG. 32 is a front elevation view of a control system;
- (34) FIG. 33 is a side elevation view of a control system;
- (35) FIG. 34 is a perspective view of a control system with part of an enclosure removed;
- (36) FIG. 35 is a top plan view of a control system with part of an enclosure removed;
- (37) FIG. 36 is a side elevation view of a control system with part of an enclosure removed;
- (38) FIG. 37 is a perspective view of an enclosure of a control system with internal components removed;
- (39) FIG. 38 is a perspective view of a circuit enclosure;
- (40) FIG. 39 is a side elevation view of a dump trailer configured to offload material to a side;
- (41) FIG. 40 is a perspective view of a dump trailer;
- (42) FIG. 41 is a perspective view of a dump trailer; and
- (43) FIG. 42 is a side elevation view of the dump trailer operating between the transporting condition and the dumping condition.
- (44) In the appended figures, similar components and/or features may have the same numerical reference label. Further, various components of the same type may be distinguished by following the reference label with a letter. If only the first numerical reference label is used in the specification, the description is applicable to any one of the similar components and/or features having the same first numerical reference label irrespective of the suffix.

DETAILED DESCRIPTION

(45) The ensuing description provides general aspects of a dump trailer and system for a semi-trailer truck only. The description is not intended to limit the scope, applicability, or configuration of the disclosure. Rather the ensuing description of the general aspects of the apparatus and system

will provide those skilled in the art with an enabling description for implementing the apparatus and/or the system. It is understood that various changes may be made in the function and arrangement of the elements without departing from the spirit and scope set forth in the appended claims. It is noted that while the following description is configured on the dump trailer and/or system for a semi-trailer truck, the system and/or components described herein may be claimed or utilized in other applications as well. For example, the trailer and/or system can be used in other trailers or specialty vehicles. Regarding other applications for the components, during the development of the overall system, various aspects were created that may be used in other trailer applications. For example, the draft arm could be an assembly, or it could be a single forged part. Another example is that the hydraulics system (sometimes referred to as a wet kit) could be retrofitted onto existing dump trailers. As such, the following description directed to a dump trailer application is meant to illustrate general aspects of the dump trailer and system for a semi-trailer truck.

(46) With reference to FIG. 1, a semi-trailer truck **10** generally includes a tractor unit **12** provided with a trailer such as the illustrated dump trailer **100**. While the semi-trailer truck **10** is most-often in a transportation condition (as illustrated in FIG. 2), occasionally the dump trailer **100** is positioned in a dump condition as illustrated in FIG. 1. During this dump condition, the weight of the dump trailer **100**, and a load **102** contained therein, are high above ground **20**. At this elevated position, the dump trailer **100** and load **102** are vulnerable to a rollover event. The ground **20** at most level-ground sites is compacted and capable of providing a stable base for the dump condition. However, certain locations are unstable. One such example location is agriculture fields during quick-turn planning events. When using the dump trailer **100** at an unstable location, the load **102** (e.g., tons of seed) may be quickly, safely, and efficiently dumped. As illustrated in FIG. 1, if rear wheels of the dump trailer **100** are positioned partially on compacted soil **22** and partially on loose-fill **24**, the dump condition can escalate to a rollover condition (or worse, a rollover event). The mechanics of this rollover event are that a tub of the dump trailer **100** is raised to cause the load **102** to slide down the tub and eject from an open end of the dump trailer **100**. In this condition, most of the weight of the dump trailer **100** and load **102** is supported by the rear wheels (where a pivot point is located). The sudden weight application onto the wheels may cause the loose-fill **24** to unexpectedly compact under the wheels. This unexpected change may result in the movement of the wheels and/or shifting which leads to the rollover condition. Traditionally, the operator of the dump process attempts to avoid the rollover event by lowering the tub as fast as possible to try to stabilize the load **102** and change the moment of inertia. Often this is successful, but occasionally the rollover condition escalates to a rollover event. When the unfortunate rollover event occurs, it is expensive and decommissions the semi-trailer truck **10** which is valuable to its operator.

(47) With reference to FIG. 2 showing a side elevation view of the semi-trailer truck **10**, the semi-trailer truck **10** most-commonly exists in the illustrated transporting condition. The dump trailer **100** is docked at its most stable location and the semi-trailer truck **10** is capable of making (short or long) trips between locations. The semi-trailer truck **10** is provided with a cable assembly **110** for power and/or information transfer between the dump trailer **100** and the tractor unit **12**. In one configuration, the cable assembly **110** is provided with a ground wire, a positive wire, a 7-wire cable, a backup air hose, and a hydraulic line. In other configurations, the cable assembly **110** is minimal and only requires a small amount of power and information to be transferred to/from the trailer **100** (e.g., a control system **800** may be provided with an on-trailer hydraulic pump, accumulator, and valves). Occasionally, the operator of the semi-trailer truck **10** desires to separate the dump trailer **100** from the tractor unit **12**, so the cable assembly **110** may be provided with a connector **112** (or multiple connectors if multiple cables exist). The connector **112** enables separation of the cable assembly **110** so a kingpin (FIG. 18, **540**, described later herein), that is the only component connecting the trailer **100** to the tractor unit **12**, can be detached from the tractor

unit **12**.

(48) With reference to FIG. **3** showing a side elevation view of a semi-trailer truck **10** in(?) the dumping condition, the dump trailer **100** may be provided with a chassis plane **30**, a tub **200**, a rear assembly **300**, a coupling plate **500**, a draft arm assembly **600**, a pair of hydraulic cylinders **700,750**, and a control system **800**. The chassis plane **30** is defined by at least two axles of the semi-trailer truck **10** that are coplanar to the chassis plane **30** as best illustrated in FIG. **3**. When configured as illustrated, the dump trailer **100** has economic benefits of at least one (or more) of the following: requiring less hydraulic oil, reducing/eliminating the need for a hydraulic system on the tractor unit **12** (which is sometimes referred to as a wet kit—an expensive and not ubiquitous subsystem, reduced manufacturing cost of a draft arm (described later herein), reduced hydraulic cylinder costs, and beneficial control of the dumping process through independent control of hydraulic cylinders that cause movement of the tub. Having provided this brief and non-limiting overview, a more detailed description will be provided.

(49) With reference to FIG. **4** showing a perspective view of a top-front side of a tub **200**, the tub **200** may be configured in any of a large number of sizes and layouts such as the illustrated U-shaped channel. The tub **200** defines an interior **202** and an exterior **204**. The tub **200** may have partitions on the interior (baffles) or be configured for a single material as illustrated. The as-illustrated tub **200** generally includes a web **210**, a left wall **220** (left side), and a right wall **230** (right side). The web **210** serves as a base for the entire tub **200** with the left wall **220** attached directly or as illustrated by a chamfered corner. Alternatively, the left wall **220** and the right wall **230** may be portions of planar or non-planar components of the dump trailer **100** (e.g., portions of a curved section of the web **210**, or physical structure extending from the tub **200**). The right wall **230** is attached to the web **210** directly or as illustrated by a chamfered corner. This U-shaped channel of the tub **200** is elongated to any reasonable length, but one example is 40 feet long (but small configurations, e.g., 21 feet, and long configurations, e.g., 60 feet, may be provided/required/configured). Additionally, the tub **200** may be constructed out of metal such as steel or aluminum and of any thickness suitable for loads and lifespan requirements (e.g., 3/16 thick steel). At one end of the tub **200** a closed end **206** may be formed by a wall of material (either one-piece or multiple-parts fastened together) and an oppositely disposed open end **208** may be capped by a movable door **209** (FIG. **3**) that can be opened automatically during a dumping event (or in a more controlled manner by mechanics).

(50) With reference to FIG. **5** showing a side elevation view of the left side of the tub **200**, the tub **200** may be provided with various structural features such as a front stiffening assembly **240**, a rear stiffening assembly **242**, a left structural beam **244**, and a right structural beam **246** (FIG. **7**). The front stiffening assembly **240** and rear stiffening assembly **242** are most commonly manufactured out of steel members (as shown) that are attached (e.g., welded) to the exterior **204** of the tub **200**. However, it should be noted that the front stiffening assembly **240** and the rear stiffening assembly **242** may be made of any of a variety of fabrication techniques or deleted as required for a particular configuration. In one of the configurations, the front stiffening assembly **240** is located approximately at a mid-way point between the open end **208** and the closed end **206** of the tub **200**; and the rear stiffening assembly **242** may be located near the open end **208** of the tub **200** due to the increasing loads applied to this general area during a dumping event. For similar reasons, the left structural beam **244** and the right structural beam **246** (FIG. **7**) may be attached to the exterior **204** of the tub **200** and span the distance from the front stiffening assembly **240** and the rear stiffening assembly **242** adjacent to (or adjoining) the web **210**.

(51) With continued reference to FIG. **5**, the tub **200** may also include a left hinge **250**, a right hinge **252** (refer to FIG. **6**), a hinge axis **260**, and a center axis **270**, and a center hinge **271** (refer to FIG. **6**). The left hinge **250** may take the form of any of a large variety of mechanisms or features capable of receiving the end of a hydraulic cylinder. One example of the left hinge **250** is a plate with a hole for receiving a yoke; another example of the left hinge **250** is a simple bolt for

receiving a ball joint (on the end of a hydraulic cylinder as described later). While the manufacturing tolerances required to properly assemble the dump trailer **100** are likely to provide enough movement, in one configuration, the use of a ball joint provides rotation and angular displacement for reasons related to adjusting the dump trailer **100** during dumping (described later herein). As shown in FIG. 5, the left hinge **250** may be attached or formed anywhere along the left wall **220** such as the illustrated location between the front stiffening assembly **240** and the closed end **206**. Furthermore, the left hinge **250** may be configured as a plate with a hole that it is welded to the exterior **204** of the tub **200**, adjacent to the left wall **220** adjoining the front stiffening assembly **240** and near a topside.

(52) With reference to FIG. 6 showing a side elevation view of the closed end **206**, the right hinge **252** may be formed on the right wall **230**. In most applications (but not necessarily all), the left hinge **250** and right hinge **252** are coaxial and generally form the hinge axis **260**. The right hinge **252** may take the form of any of a large variety of mechanisms or features capable of receiving the end of a hydraulic cylinder. One example of the right hinge **252** is a plate with a hole for receiving a yoke; another example of the right hinge **252** is a simple bolt for receiving a ball joint (on the end of a hydraulic cylinder as described later). While the manufacturing tolerances required to properly assemble the dump trailer **100** are likely provide enough movement, in one configuration, the use of a ball joint provides rotation and angular displacement for reasons related to adjusting the dump trailer **100** during dumping (described later herein). As also shown in FIG. 6, the right hinge **252** may be attached or formed anywhere along the right wall **230** such as the illustrated location between the front stiffening assembly **240** and the closed end **206** (FIG. 5). Furthermore, the right hinge **252** may be configured as a plate with a hole that it is welded to the exterior **204** of the tub **200**, adjacent to the right wall **230** and adjoining the front stiffening assembly **240**.

(53) With continued reference to FIG. 6, the tub **200** may further include a center axis **270** concentric to a center hinge **271** configured as one single hinge member. The center hinge **271** may be a simple pivot point or, as illustrated, as pair of hinges such a center-left hinge **272** and a center-right hinge **274**. FIG. 6 also shows that the tub **200** may include a center plane **280**. The center plane **280** is generally located between the left wall **220** and the right wall **230** in a symmetrical configuration (sometime referred to herein as symmetrically formed).

(54) With reference to FIG. 7 showing a bottom plan view of the tub **200**, the center-left hinge **272** may be attached to the left structural beam **244** and the right center hinge **274** may be attached to the right structural beam **246**. The center hinge **271** (either as a single feature or the pair including the center-left hinge **272** and the center-right hinge **274**) form the center axis **270**. The center axis **270** is generally formed parallel to the hinge axis **260** for purposes of controlling movement of the tub **200** as described herein. With continued reference to FIG. 7, the center plane **280** is symmetrical to the left wall **220** and the right wall **230** whereby the hinge axis **260** and center axis **270** pierce through the center plane **280**.

(55) With reference to FIG. 8 showing a top perspective view of a rear assembly **300**, the rear assembly **300** may be provided with a front axle assembly **310**, a rear axle assembly **330**, a left spring assembly **350**, a right spring assembly **390**, a pyramid **430**, and a rear pivot **450** (refer to FIG. 14).

(56) With continued reference to FIG. 8, the front axle assembly **310** includes an axle **312**, a left wheel **314**, a right wheel **316**, a left damper **318**, and a right damper **320**. The left wheel **314** and right wheel **316** are attached to the axle **312** in a manner well known in the art, specifically with hubs that may (or may not include) brakes. If configured with air springs as illustrated in FIG. 8, the left damper **318** and the right damper **320** are attached to the leading side of the axle **312** and are provided with a detailed 4-bar mechanism that is known in the art. The axle **312** may also include a left spring contact **322** and a right spring contact **324**, both formed on the axle **312** for receiving spring assemblies (e.g., left spring assembly **350** and right spring assembly **390**).

(57) With reference to FIG. 9 showing a bottom perspective view of the rear assembly **300**, the rear

axle assembly **330** includes an axle **332**, a left wheel **334**, a right wheel **336**, a left damper **338** and a right damper **340**. The left wheel **334** and right wheel **336** are attached to the axle **332** in a manner as described with the front axle assembly **310**. If configured with air springs, the left damper **338** and the right damper **340** are attached to the leading side of the axle **332**. The axle **332** may also include a left spring contact **342** and a right spring contact **344**, both formed on the axle **332** for receiving spring assemblies.

(58) With reference to FIG. **10** showing a perspective view of the left spring assembly **350**, the left spring assembly **350** may take any of a variety of configurations such as the illustrated configuration including a first axle attachment **352**, a pair of u-bolts **354**, a main leaf spring **356**, a second leaf spring **358**, and a third leaf spring **360**. The main leaf spring **356** has a first end **362** and an oppositely disposed second end **364** with a midpoint **366**. At the first end **362** of the main leaf spring **356**, the plurality of individual springs **356**, **358**, **360** are held together by the first axle attachment **352** and the u-bolts **354**. At the second end **364** of the main leaf spring **356**, the left spring assembly **350** may also include a second axle attachment **368** and a pair of u-bolts **370** capture the springs **356**, **358**, **360** as illustrated. At the midpoint **366** of the main leaf spring **356**, the left spring assembly **350** may be provided with hinge assembly **372** and a pair of u-bolts **374**. The hinge assembly **372** adjoins the main leaf spring **356** and the u-bolts **374** capture the second leaf spring **358** and third leaf spring **360**.

(59) With reference to FIG. **11** showing a side view of the left spring assembly **350**, the left spring assembly **350** includes an axle plane **376**. The axle plane **376** is defined by the axes (plural of axis) of rotation of the axle **312** of the front axle assembly **310** and the axle **332** of the rear axle assembly **330**, both of the center of rotations of the axles **312** and **332** reside in the axle plane **376**, as illustrated in FIG. **11**. The hinge assembly **372** includes a pivot axis **378** about which the tub **200** rotates. There may be a left deflection distance **380** defined as the distance between the pivot axis **378** and the axle plane **376**.

(60) With reference to FIG. **12** showing a perspective view of a right spring assembly **390**, the right spring assembly **390** may take any of a variety of configurations such as the illustrated configuration including a first axle attachment **392**, a pair of u-bolts **394**, a main leaf spring **396**, a second leaf spring **398**, and a third leaf spring **400**. The main leaf spring **396** has a first end **402** and an oppositely disposed second end **404** with a midpoint **406**. At the first end **402** of the main leaf spring **396**, the plurality of individual springs **396**, **398**, and **400** are held together by the first axle attachment **392** and the u-bolts **394**. At the second end **404** of the main leaf spring **396**, the right spring assembly **390** may further include a second axle attachment **408** and a pair of u-bolts **410** capture the springs **396**, **398**, and **400** as illustrated. At the midpoint **406** of the main leaf spring **396**, the right spring assembly **390** may be provided with hinge assembly **412** and a pair of u-bolts **414**. The hinge assembly **412** adjoins the main leaf spring **396** and the u-bolts **414** capture the second leaf spring **398** and the third leaf spring **400**.

(61) With reference to FIG. **13** showing a side view of the right spring assembly **390**, the right spring assembly **390** includes an axle plane **416**. The axle plane **416** is defined by the axes (plural of axis) of the axle **312** of the front axle assembly **310** and the axle **332** of the rear axle assembly **330**, both of the center of rotations of the axles **312**, **332** reside in the axle plane **416** as illustrated in FIG. **13**. The hinge assembly **412** includes a pivot axis **418** about which the tub **200** rotates. There is a left deflection distance **420** defined as the distance between the pivot axis **418** and the axle plane **416**.

(62) With reference to FIG. **14** showing a side elevation view of the rear assembly **300**, the pyramid **430** may be configured as illustrated with a top **432**, a bottom **434**, a front **436**, a back **438**, a left **440**, and a right **442** (FIG. **8**) to generally form an inverted pyramid shape. The top **432**, the bottom **434**, the front **436**, the back **438**, the left **440**, and the right **442** may be constructed out of a variety of materials and/or manufacturing processes, such as out of cold-rolled sheet steel that is welded into a shape defining an interior portion **444** and an exterior portion **446**. The interior portion **444** is

often configured with various mechanical items such as components of a braking system or other sub-systems that need a protected location. The top **432** of the pyramid **430** may be attached to the tub **200** in any of a variety of manners, such as by welding. If, for example, the pyramid **430** is welded to the tub **200**, the left **440** is welded to the left structural beam **244** (FIG. 7) near the rear stiffening assembly **242**. This illustrated configuration would also result in the right **442** being welded to the right structural beam **246** (FIG. 7).

(63) With continued reference to FIG. 14, the rear assembly **300** is further provided with a rear pivot **450**. The rear pivot **450** may be a solid or hollow circular profile, such as the illustrated solid bar of steel. The rear pivot **450** is installed as illustrated so that the rear pivot **450** is attached to the hinge assembly **372** and the hinge assembly **412** (FIG. 12) to provide an axis of rotation about which the tub **200** (and intermediary pyramid **430**) pivots.

(64) With reference to FIG. 15 showing a perspective view of the bottom **434** of the pyramid **430** with the left wheel **334** of the rear axle assembly **330** removed, the rear assembly **300** is further provided with provisions for attaching the pyramid **430** to the rear pivot **450**, such as left pyramid mount **460** and right pyramid mount **470**. The left pyramid mount **460** is attached to the bottom **434** near the left **440** of the pyramid **430**. The right pyramid mount **470** is attached to the bottom **434** near the left **440** of the pyramid **430**. The left pyramid mount **460** is provided with a hinge such as a hole **462**; and the right pyramid mount **470** is provided with a hinge such as a hole **472**. When assembled as best illustrated in FIG. 15, the hole **462** of the left pyramid mount **460** is coaxial to the rear pivot **450** and capable of forming an axis of rotation about which the tub **200** may rotate relative to the rear assembly **300**. Furthermore, the hole **472** of the right pyramid mount **470** is coaxial to the hole **462** of the left pyramid mount **460** and the rear pivot **450**.

(65) With reference to FIG. 16 showing a rear elevation view of the rear assembly **300** wherein the ground **20** is not level, the components of the rear assembly **300** cooperate to provide independent movement of the tub **200** relative to the rear assembly **300**. This independent movement is best illustrated by the left deflection distance **380** of the left spring assembly **350** and the left deflection distance **420** of the right spring assembly **390** in a levelling process best illustrated by keeping rear pivot **450** 'level' to the ground **20**. This leveling process is described herein, but as iterated, one of the purposes is to keep the tub **200** and its load perpendicular to the ground **20** as illustrated by the center plane **280** of the tub **200** as shown. As shown in FIG. 16, the center plane **280** is not perpendicular to the ground **20**, therefore the tub **200** is in a canted condition wherein the load held in the tub **200** is unstable when in the dumping condition.

(66) With reference to FIG. 17 showing a perspective view of a coupling plate **500**, the coupling plate **500** may take a number of forms, but one particular form is illustrated as a cuboid shape with a top-surface removed. In this illustrative example, the coupling plate **500** generally defines an interior **502** and an exterior **504** including a base **510**, a left side **512**, a right side **514**, front side **516** and a back side **518**. The left side **512**, right side **514**, front side **516**, and back side **518** may be integrally formed with the base **510** (e.g., bent out of a flat material) or welded into the illustrated configuration. The coupling plate **500** may have additional features such as: partitions, features, bearing surfaces, brackets, and cable routing features. Additional features illustrated in FIG. 17 include: a left hole **520**, a right hole **522**, a left hydraulic cylinder clearance **524**, a right hydraulic cylinder clearance **526**, and a draft arm clearance **528**. The left hole **520** is formed in the left side **512** and the right hole **522** is formed in the right side **514** to form a coupling axis **530**. The left hole **520** and the right hole **522** are coaxial to the coupling axis **530**.

(67) With reference to FIG. 18 showing a rear elevation view of the coupling plate **500**, the coupling plate **500** is provided with a feature for attaching the dump trailer **100** to the semi-trailer truck **10** such as a kingpin **540**. The kingpin **540** is formed on the exterior **504** of the base **510**. The kingpin **540** is common in the industry and serves to make the dump trailer **100** readily and removably attached to the semi-trailer truck **10** to enable different configurations of the semi-trailer truck **10**.

(68) With reference to FIG. 19 showing a front top perspective view of a draft arm assembly 600, the draft arm assembly 600 may take a variety of configurations including a tube 610 with a rectangular profile (e.g., a steel tube). The tube 610 includes a top side 612, a bottom side 614, a left side 616, a right side 618, a proximal end 620, and a distal end 622. The sides 612, 614, 616, and 618 may create an elongated tube capable of defining a protected interior portion 624 for receiving cables or other components (e.g., cable assembly 110). The draft arm assembly 600 is provided with a proximal hinge 630 and a distal hinge 640 located at the proximal end 620 and the distal end 622, respectively. The proximal hinge 630 and distal hinge 640 may take any of a number of forms, but in the illustrated configuration are configured as a proximal tube 632 and a distal tube 642 capable of receiving individual pins/rods as described later herein. The proximal tube 632 passes through the left side 616 and the right side 618 of the steel tube 610. The distal tube 642 passes through the left side 616 and the right side 618 near the distal end 622.

(69) With reference to FIG. 20 showing a top plan view of the draft arm assembly 600, the proximal hinge 630 defines a draft arm proximal axis 634 and the distal hinge 640 defines a draft arm distal axis 644.

(70) With reference to FIG. 21 showing a perspective view of a left hydraulic cylinder 700 in an exploded condition, the left hydraulic cylinder 700 may be any of a large variety of hydraulic cylinders, such as the illustrated single-stage cylinder. With the single-stage cylinder, a compressed length is roughly half of a fully-extended length. In the illustrated configuration, the compressed length is about 13 feet and the extended length is about 20 feet. The left hydraulic cylinder 700 is provided with a body 710 that includes a cap end head 712 and a rod end head 714. The cap end head 712 is provided with an attachment feature such as hole 716 (as illustrated). The rod end head 714 is provided with a sealing feature configured to receive a piston rod 720. The piston rod 720 is provided with a piston end 722 for sealing against the body 710 and creating a coupling-side cavity 730 disposed between the cap end head 712 and the piston end 722 of the piston rod 720. The left hydraulic cylinder 700 may also be provided with a rod-side cavity 732 disposed between the rod end head 714 of the body 710 and the piston end 722 of the piston rod 720. The coupling-side cavity 730 and the rod-side cavity 732 change volumes depending on the position of the piston rod 720 as it extends from and retracts into the body 710. In order to move fluid, almost exclusively a liquid (e.g., hydraulic oil), into and out of the cavities 730, 732, the left hydraulic cylinder 700 is provided with fittings such as a coupling-side fitting 734 and a rod-side fitting 736. The locations and existing of these fittings can be as illustrated in FIG. 21 or moved to different locations on the left hydraulic cylinder 700 (e.g., the coupling-side fitting 734 may be located at the end of the piston rod 720 whereby the length of the piston rod 720 acts as a supply line for the fluid).

(71) With continued reference to FIG. 21, the left hydraulic cylinder 700 is provided with an attachment hinge 724 formed on the piston rod 720 at a location oppositely disposed from the piston end 722. The attachment hinge 724 may take the form of any of a variety of connection mechanisms used in industry, such as hole 726 as illustrated. The attachment hinge 724 may be a rigid attachment wherein the left hydraulic cylinder 700 remains perpendicular to the attachment point, or it may have a floating configuration wherein the left hydraulic cylinder 700 may move a few degrees from a perpendicular orientation relative to the object it is attached to (e.g., the tub 200 as described herein).

(72) With reference to FIG. 22 showing an assembled condition of the left hydraulic cylinder 700, the left hydraulic cylinder 700 includes a proximal hinge axis 702 and a distal hinge axis 704. The proximal hinge axis 702 is coaxial to the hole 716 formed in the cap end head 712. The distal hinge axis 704 is coaxial to the hole 726 formed in the attachment hinge 724. In most situations, the proximal hinge axis 702 is parallel to the distal hinge axis 704 and separated by a left hydraulic cylinder length 706 that is variable. In a load leveling event, the distal hinge axis 704 and the proximal hinge axis 702 may be slightly non-parallel in order to adjust to variable ground as will be described later herein.

(73) With reference to FIG. 23 showing a perspective view of a right hydraulic cylinder 750 in an exploded condition, the right hydraulic cylinder 750 may be similar to the right hydraulic cylinder 700. The right hydraulic cylinder 750 is provided with a body 760 that includes a cap end head 762 and a rod end head 764. The cap end head 762 is provided with an attachment feature such as the hole 766. The rod end head 764 is provided with a sealing feature configured to receive a piston rod 770. The piston rod 770 is provided with a piston end 772 for sealing against the body 760 and creating a coupling-side cavity 780 disposed between the cap end head 762 and the piston end 772 of the piston rod 770. The right hydraulic cylinder 750 may also be provided with a rod-side cavity 782 disposed between the rod end head 764 of the body 760 and the piston end 772 of the piston rod 770. The coupling-side cavity 780 and the rod-side cavity 782 change volumes depending on the position of the piston rod 770 as it extends from and retracts into the body 760. In order to move fluid, almost exclusively a liquid (e.g., hydraulic oil), into and out of the cavities 780 and 782, the right hydraulic cylinder 750 is provided with fittings such as a coupling-side fitting 784 and a rod-side fitting 786. The locations and existing of these fittings can be as illustrated in FIG. 23 or moved to different locations on the right hydraulic cylinder 750 (e.g., the coupling-side fitting 784 may be located at the end of the piston rod 770 whereby the length of the piston rod 770 acts as a supply line for the fluid).

(74) With continued reference to FIG. 23, the right hydraulic cylinder 750 is provided with an attachment hinge 774 formed on the piston rod 770 at a location oppositely disposed from the piston end 772. The attachment hinge 774 may take the form of any of a variety of connection mechanisms used in the industry, such as a hole 776 as illustrated. The attachment hinge 774 may be a rigid attachment wherein the right hydraulic cylinder 750 remains perpendicular to the attachment point, or it may have a floating configuration wherein the right hydraulic cylinder 750 may move a few degrees from a perpendicular orientation relative to the object it is attached to (e.g., the tub 200 as described herein).

(75) With reference to FIG. 24 showing an assembled condition of the right hydraulic cylinder 750, the right hydraulic cylinder 750 includes a proximal hinge axis 752 and a distal hinge axis 754. The proximal hinge axis 752 is coaxial to the hole 766 formed in the cap end head 762. The distal hinge axis 754 is coaxial to the hole 776 formed in the attachment hinge 774. In most situations, the proximal hinge axis 752 is parallel to the distal hinge axis 754 and separated by a right hydraulic cylinder length 756 that is variable. In a load leveling event, the distal hinge axis 754 and the proximal hinge axis 752 may be slightly non-parallel in order to adjust to variable ground as will be described later herein.

(76) With reference to FIG. 25 showing a bottom front perspective view of the semi-trailer truck 10 and one illustrative configuration of a control system 800, the control system 800 may include a hydraulic pump 810, an accumulator 820, a controller 830, and a plurality of hydraulic lines 840 (e.g., a pump-to-accumulator line 842, an accumulator-to-left-valve line 844, an accumulator-to-right-valve line 846), a left control valve 850, and a right control valve 860. While these components are listed, it is specifically understood that some or all of these may be eliminated or relocated (e.g., to the tractor unit 12) and the scope of the present illustrated system. The control system 800 is shown on the exterior 204 of the tub 200, specifically on the exterior of the web 210 for illustrative purpose only. These components may be enclosed in a housing or somewhere protected on the dump trailer 100 (e.g., inside the pyramid 430). The hydraulic pump 810 may be any of a variety of pumps configured to pressurized fluid (specifically liquid and most likely a hydraulic oil). The hydraulic pump 810 only has so much capacity therein, so the pump-to-accumulator line 842 is provided for conveying the pressurized fluid from the hydraulic pump 810 to the accumulator 820 where a variable amount of fluid can be stored. Ideally, this accumulator 820 will be configured so components powered by pressurized fluid therein can be quickly transferred without waiting for the hydraulic pump 810 to do all of the pressurization on-demand (e.g., with about 10 to 50 gallons of hydraulic oil). Rather, the pressurization is completed during a

period of time leading up to the call for pressurized fluid. The pressurized fluid may be transferred from the accumulator **820** to the left hydraulic cylinder **700** and the right hydraulic cylinder **750** via lines. In one configuration, a pair of hydraulic lines referred to herein as the accumulator-to-left-valve line **844** and the accumulator-to-right-valve line **846** transfer the fluid to the left control valve **850** and the right control valve **860**, respectively. The control system **800** further includes a left supply line **852** in fluid communication between the left control valve **850** and the left hydraulic cylinder **700**. The control system **800** further includes a right supply line **862** in fluid communication between the right control valve **860** and the right hydraulic cylinder **750**. While one line of fluid communication is described to each of the hydraulic cylinders, it is understood that some types of hydraulic cylinders are configured with bi-directional operation characteristics that require two separate supply lines for providing fluid to act on opposite sides of pistons in the hydraulic cylinders. If required, a bi-directional configuration could be implemented in a similar manner as previously described.

(77) With further reference to FIG. 25, the controller **830** of the control system **800** may be provided with a means for sensing orientation of the tub **200**. While a large variety of mechanism exist for sensing orientation, one mechanism is an accelerometer **832** capable of providing side-to-side angle of the tub **200** (gravitational force of earth, e.g., plumb/true). The accelerometer **832** may be configured with other sensors such as an angle-of-inclination of the tub **200** that reports back the current angle of the tub **200**. While less likely to be utilized, the accelerometer **832** may be configured with a third-axis of reported angle which is referred to herein as a directional reading. The controller **830** is further provided with an electrical supply line **834** to transfer power to the hydraulic pump **810**. The electrical supply line **834** is capable of transferring power sourced from the tractor unit **12** to the hydraulic pump **810** when there is a need for increasing pressure of the fluid. Additionally, the controller **830** is provided with a power lead **836** capable of sourcing the power from an on-trailer power supply (e.g., a battery or a generator) or directly from the tractor unit **12**. Briefly summarized, during normal operation, power is delivered from the tractor unit **12** via the cable assembly **110** through the power lead **836** through the controller **830** and to the hydraulic pump **810** via the electrical supply line **834**. Fluid is pressurized by the hydraulic pump **810** and transferred to the accumulator **820** via the pump-to-accumulator line **842** where it further travels to the left hydraulic cylinder **700** via the left supply line **852**, the left control valve **850**, and the accumulator-to-left-valve line **844**. In a similar manner, the fluid travels to the right hydraulic cylinder **750** via the right supply line **862**, the right control valve **860**, and the accumulator-to-right-valve line **846**. In some situations, the operation of the left hydraulic cylinder **700** and the right hydraulic cylinder **750** are synchronized (wherein the displacement and speed are matched). In other situations, the operation of the left hydraulic cylinder **700** and the right hydraulic cylinder **750** are independent (e.g., when loose-fill ground is located at a site and the accelerometer **832** senses and reports an angle-change to the controller **830**).

(78) With continued reference to FIG. 25, the draft arm assembly **600** is attached to the tub **200** as illustrated by a pivot rod **602** capturing the distal tube **642** of the distal hinge **640** (both in FIG. 19). This capturing enables the draft arm assembly **600** to be rotatably attached to the tub **200** about the center axis **270** (FIG. 6) because the draft arm distal axis **644** (FIGS. 19 and 20) is coaxial to the center axis **270**. This positions the draft arm assembly **600** between the center-right hinge **274** and the center-left hinge **272** (both FIG. 6).

(79) With reference to FIG. 26 showing a side perspective view of the semi-trailer truck **10**, the draft arm assembly **600** is attached to the coupling plate **500** by a control pin **506** as illustrated. This attachment results in the proximal end **620** of the draft arm assembly **600** being attached to the coupling plate **500** between the left side **512** and the right side **514** of the coupling plate **500**. The control pin **506** passes through the left hole **520** (FIG. 17) and the right hole **522** (FIG. 17) to create a hinge coaxial to the coupling axis **530** about which the draft arm assembly **600** is rotationally attached via the proximal hinge **630** (of the draft arm assembly **600**). This places the draft arm

proximal axis **634** (FIG. **20**) coaxial to the coupling axis **530**. It is important to note that in the transportation condition best illustrated in FIG. **2**, the draft arm assembly **600** is positioned to be slightly non-parallel to the ground **20**. With reference to FIG. **2**, the draft arm assembly **600** intersects a ground plane of the ground **20** at draft arm angle **604** that is configured to always cause the linkage of the draft arm assembly **600** to move the closed end **206** of the tub **200** away from the ground **20**. If the draft arm angle **604** was zero or slightly negative, the tub **200** would rotate towards ground **20** and therefore be inoperable. Therefore, the draft arm distal axis **644** (which is also coaxial to the center axis **270**) is slightly 'above' the draft arm proximal axis **634** (which is also coaxial to the coupling axis **530**) because any other orientation that places the draft arm distal axis **644** at the same level or slightly below the draft arm proximal axis **634** would bind the mechanism and it would not function as intended. In other words, the draft arm assembly **600** is non-parallel to the ground (and/or a plane defined by the coupling plate).

(80) With reference again to FIG. **21**, the hole **716** formed in the cap end head **712** of the body **710** on the left hydraulic cylinder **700** is captured by the control pin **506**. This results in the left hydraulic cylinder **700** being rotatably coupled to the coupling plate **500** about the coupling axis **530** (and the coaxial proximal hinge axis **702**, FIG. **22**). The draft arm assembly **600** may, typically, be oriented at the midpoint of the control pin **506** while the left hydraulic cylinder **700** is oriented by the left side **512** of the coupling plate **500**. As further illustrated in FIG. **26**, the hole **766** (FIG. **24**) formed in the cap end head **762** of the body **760** on the right hydraulic cylinder **750** is captured by the control pin **506**. This results in the right hydraulic cylinder **750** being rotatably coupled to the coupling plate **500** about the coupling axis **530** (and the coaxial proximal hinge axis **752**, FIG. **24**). The draft arm assembly **600** may, typically, be oriented at the midpoint of the control pin **506** while the right hydraulic cylinder **750** is oriented by the right side **514** of the coupling plate **500**. The hole **776** (FIG. **24**) formed on the attachment hinge **774** positioned on the piston rod **770** of the right hydraulic cylinder **750** is adjacent to and coaxial with the right hinge **252** formed on the tub **200**. This places the distal hinge axis **754** (FIG. **24**) coaxial with the hinge axis **260** (FIG. **6**) whereby the right hydraulic cylinder **750** is rotatably attached to the tub **200**.

(81) With reference to FIG. **27** showing a front left side perspective view of the semi-trailer truck **10**, the left hydraulic cylinder **700** is rotatably attached to the tub **200** via the hole **726** formed on the attachment hinge **724** positioned on the piston rod **720** of the left hydraulic cylinder **700**. This orientation results in the left hinge **250** formed on the tub **200** (refer to FIG. **6**) being adjacent to and coaxial with the hole **726** (refer to FIG. **22**). This places the distal hinge axis **704** (FIG. **22**) coaxial with the hinge axis **260** (FIG. **6**) whereby the left hydraulic cylinder **700** is rotatably attached to the tub **200**.

(82) With reference to FIG. **28** showing a side elevation view of the semi-trailer truck **10** with the movement from the transporting condition to the dumping condition, the control system **800** implements a process for controlling expansion of the left hydraulic cylinder **700** and the right hydraulic cylinder **750** to cause the closed end **206** of the tub **200** to move away from ground **20** in a controlled and improved manner. To enable this controlled expansion, the control system **800** senses any subpar conditions that may lead to a rollover condition and/or a probability of a rollover event. In response to sensing, the control system **800** may be pre-programmed to take any of a number of actions such as: moving the semi-trailer truck **10** forward or backwards; slowly lowering either or both of the left hydraulic cylinder **700** and right hydraulic cylinder **750**; quickly lowering either or both of the left hydraulic cylinder **700** and right hydraulic cylinder **750**; slowly raising either or both of the left hydraulic cylinder **700** and right hydraulic cylinder **750**; quickly raising either or both of the left hydraulic cylinder **700** and right hydraulic cylinder **750**; or other actions implied and feasibly but not specifically described herein. If the left hydraulic cylinder **700** is operated separately from the right hydraulic cylinder **750**, the tub **200** is shifted to cause asymmetrical movement of the left spring assembly **350** and right spring assembly **390** of the rear assembly **300** (all shown in FIG. **8**) to cause changes of the left deflection distance **380** and left

deflection distance **420** (both shown in FIG. **16**).

(83) By operating the left hydraulic cylinder **700** and right hydraulic cylinder **750** independently, the control system **800** can greatly improve the safety and therefore speed of the dumping condition. In accomplishing this, the system improves the speed of movement of load(s) from a first location to a second location (where the dumping event occurs).

(84) In one alternative configuration, the dump trailer **100** may be provided with linear measurement equipment on the left hydraulic cylinder **700** and/or the right hydraulic cylinder **750**. Examples of linear measurement equipment include, for example: laser, friction wheel, linear encoder, etc. This measurement equipment may provide a feedback loop to the control system **800** for ensuring that the system is properly operating as intended and nothing is malfunctioning (due to normal wear, adverse operation, and/or improper use of the dump trailer **100**).

(85) In one alternative configuration, the accelerometer **832** can be on a main circuit board of the controller **830**. Alternatively, the accelerometer **832** may be mounted at a different location and in communication with the control system **800** either wired or wirelessly. Furthermore, the accelerometer **832** may be similar to Freescale's MMA8452Q which is low-power, three-axis, capacitive micro-machined accelerometer with 12 bits of resolution. The accelerometer **832** is able to sense orientation or motion and to report it back to a microcontroller (for example) via a standard communication protocol (e.g., I2C, SPI).

(86) In one alternative configuration, a mobile device and/or an application running on a device can track the efficacy of the dumping event. This application can measure, track and ranking the safety record of the site, the truck, the operator of the truck individually and statistically. The application can report this safety record to at least one of the following to a regulatory body, a truck owner, and/or an insurance company. The dump trailer **100** may include a control system with an application-based interface such as a mobile device (e.g., cellphone, tablet, portable media player, radio deck with interface features, etc.). In this alternative configuration, the phone provides metrics about the quality-of-safety for the transportation process (including, for example, the loading and unloading processes). The device that the application is running on may include a spectrum chip operable with the control system for communicating under various protocols (e.g., Bluetooth, WiFi, cellular, 5G, 4G, CDMA, GSM, etc.) to external machines (e.g., portable devices, signals, servers, the internet, etc.).

(87) In one alternative configuration, the rear assembly **300** may be configured with a single axle having at least a pair of left and right wheels. The springs can be coil-over springs or special tires capable of deflecting may be utilized in replacement of the more complex configuration of FIG. **8**.

(88) In one alternative configuration, the left wall, right wall, and web of the tub **200** can be formed from out of single sheet of material (either folded and welded, pieced together, or slip-rolled if a circumferential profile).

(89) In one alternative configuration, the control system **800** may be positioned inside the pyramid **430** where it is protected from debris and impact.

(90) In one alternative configuration, the rear assembly **300** may be configured with a single axle having at least a pair of wheels and air springs that replace the more complex spring assembly.

(91) In one alternative configuration illustrated in FIG. **27**, a left cylinder **900** may be configured to act as a double-acting cylinder but modified so that one region is a reservoir for storing and dispensing hydraulic fluid. As best illustrated in FIG. **27**, the left cylinder **900** includes a partition **902**, a working end **910**, a first port **912**, a second port **914**, a reservoir end **920**, a third port **922**, a fourth port **924** and a working distance **930**. The partition **902** separates the working end **910** from the reservoir end **920**; for example, a traditional long hydraulic cylinder could be modified with a baffle inserted and sealingly engaged to the interior portion of the left cylinder **900**. In this configuration, the first port **912** and the second port **914** are in fluid communication with the working end **910** for moving the rod relative to the bore. To extend the rod from the bore (thereby expanding the left cylinder **900**), hydraulic fluid is introduced to the second port **914** and to shorten

the left cylinder **900**, hydraulic fluid is removed from the second port **914** and hydraulic fluid is moved into the first port **912**. This alternative configuration has the reservoir end **920** where hydraulic fluid can be stored. Either air or hydraulic fluid can be added or removed via the third port **922** and the fourth port **924**. One particular feature of this configuration is that the fourth port **924** is generally the lowest part of the configuration so hydraulic fluid is readily sourced from the fourth port **924**. It should be understood by those skilled in the art that various components of the control system **800** may be in fluid communication with the left cylinder **900** via one or more of the first port **912**, the second port **914**, the third port **922**, and the fourth port **924**. It should be further understood that the teachings of the left cylinder **900** may be present in a right cylinder **950**, but details thereof are not described herein. It is to be understood that the left cylinder **900** and the right cylinder could also be configured as a single-acting cylinder (e.g., the second port **914** may simply be open to air or outright deleted).

(92) In one alternative configuration, the left cylinder **900** may be provided with pressure sensors (e.g., left pressure sensor **980**) configured to measure pressure of the hydraulic fluid. In this configuration, the pressure sensors may be attached to each of the hydraulic cylinders and interfaced with a control system where the pressures can be compared. If a threshold (either predetermined, spot-determined, adapted, or otherwise indicated) between the left cylinder pressure and the right cylinder pressure is exceeded, the system may be halted (e.g., stopped) and/or the tub may be lowered until the excessive difference of pressure is reduced or eliminated. It should be noted that the pressure may be monitored anywhere that is in fluid communication. For example, there may be a manifold from which the various hydraulic lines are connected and therefore, the left pressure sensor **980** may be attached to a manifold (not shown) and generally interfaced with the control system **800**.

(93) With reference to FIG. **29** showing a side elevation view of an illustrative dump trailer **100** provided with a control system **2900** (sometimes also referred to herein as control system **800**). As shown in the illustrative configuration, the control system **2900** may be an enclosure fully containing (most if not all) of the components required for operating the dump trailer **100**. As disclosed in paragraph 0070 herein (and other patent applications by the same inventorship), the control system **2900** may receive power through the electrical supply line **834** transferring power sourced from the tractor unit **12** (FIG. **1**) to the control system **2900**. Additionally, the control system may be provided with a power lead capable of sourcing power from an on-trailer power supply such as battery or a generator (in cooperation with or entirely separate from power sourced directly from the tractor unit **12**).

(94) With reference to FIG. **30** showing an isometric perspective of one illustrative configuration of the control system **2900** removed from the dump trailer **100**. While the control system **2900** may be attached in a number of ways and/or orientations, the following description will benefit from an orientation wherein the control system **2900** may form an enclosure **2902** has a top **2910**, a bottom **2912**, a left **2914**, a right **2916**, a front **2918**, and a back **2920** configured in a generally parallelepiped geometry as illustrated. The material of the enclosure **2902** may be ingress-repellant or ingress-proof and/or capable of withstanding the road conditions (e.g., salt water environments or winter conditions with traction-aiding stones/sand). In one configuration, the enclosure **2902** is a stainless steel that is folded into the configuration, welded where appropriate, and/or configured with gaskets (e.g., where serviceman access is required). The enclosure **2902** may have various features attached thereto such as a circuit enclosure **2930** and a fan shroud **2932**. The circuit enclosure **2930** may, for example, be attached to the front **2918** of the enclosure **2902**. The fan shroud **2932** may also be attached to the front **2918** with an opening **2934** configured where water is less-than-likely to enter the main enclosure **2902**.

(95) With reference to FIG. **31** showing plan view of the top **2910** of the control system **2900**. The enclosure **2902** may be provided with a second fan shroud **2936** formed on the back **2920**. If configured with fan shrouds (e.g., one or both of the fan shrouds **2932**, **2936**), they may create a

cooling flow path **2938**. The cooling flow path **2938** may move air therethrough by convective forces or via mechanized air handling assistance (e.g., one or more fans as described elsewhere herein).

(96) With reference to FIG. **32** show side elevation view of the top **2910**, the control system **2900** may also include a plurality of connections such as hydraulic inlet port **2940** and hydraulic outlet port **2942** formed in the front **2918**. The top **2910** may also include a plurality of cord grommets formed in the circuit enclosure **2930** such as inlet gromet **2946** and outlet gromet **2948**. The gromets **2946** and **2948** may enable cables such as sensor and power cables to enter the circuit enclosure **2930** and/or all higher-current carrying cables such as solenoid wires to exit the outlet gromet **2948**. It should be understood that the location and purpose of these various gromets is provided for illustrative purposes only and may be relocated or altered depending on the particular configuration.

(97) With reference to FIG. **33** showing an elevation view of the right **2916** side of the control system **2900**, the fan shrouds **2936** may be provided with an opening **2950** formed similar to the opening **2934** in the fan shroud **2932**. When configured with the openings **2934** and **2950**, the cooling flow path **2938** has a clear path through the enclosure **2902**. The circuit enclosure **2930** may include a circuit board assembly **2960**. The circuit board assembly **2960** may be configured components and features to enable proper configuration and operation of the control system **2900**. For example, the circuit board assembly **2960** may be provided with a positioning module **2962**. The positioning module **2962** may be any of a variety of geolocation or relative position devices such as a multiband geolocation positioning system (GPS). The positioning module **2962** may have an integrated multiband technology for centimeter level accuracy to enable precise location of the control system **2900** (and components attached thereto such as the dump trailer **100**). The positioning module **2962** can receive and track multiple GNSS systems such as the four major GNSS constellations (GPS, Galileo, GLONASS and BeiDou) through an on-board or remote antenna (not shown). Separately, but somewhat related, the circuit board assembly **2960** of the control system **2900** may further include a sensor intake **2964** for intaking and/or processing signal data from comments of the dump trailer **100** such as pressure transducers (fluid and/or load) as well as forces and displacements. The circuit board assembly **2960** may further include near-range or long-range communications devices such as a telemetry device **2966**. If provided with the telemetry device **2966**, the control system **2900** is able to send and/or receive signals such as status, location, adherence to a safety protocol, authorization for operation, max speed, average speed, propensity/occurrence of near-rollover event(s), etc.

(98) With reference to FIG. **34** showing perspective view of the control system **2900** with the top **2910** removed therefrom (not shown). The control system **2900** may be provided with a battery management system **2970**, a battery **2972**, a shunt **2974**, a motor controller **2976**, a motor **2978**, and a pump **2980**. In one configuration, the battery management system **2970** may buck or boost a supply voltage to properly match optimized charging voltage(s) of the battery **2972**. The **2970** may pulse power delivery to the battery **2972**, or altogether terminate delivery of the power to the battery **2972**. Furthermore, the **2970** may be configured to prohibit the transfer of power from the **2972** to other equipment 'upstream' from the control system **2900** (e.g., if a charging power supply is not active, power in the battery **2972** should not flow to the cab via the cable assembly **110**, in one instance). The battery **2972** may be configured as a single cell or a plurality of cells/batteries such as the illustrated configuration as four batteries in series resulting in 48-volt nominal rating. The battery **2972** may be configured to deliver power to the motor **2978** through a sequential delivery through the shunt **2974** and the motor controller **2976**. The shunt **2974** may enable the measurement of power and/or current from the battery **2972** to the motor **2978**. Alternatively, the shunt **2974** may be replaced with a switch capable of removing power delivery from the battery **2972**. The motor controller **2976** may be any of a variety of devices including one, some, or all of the following features: phase converter, controller with Bluetooth telemetry, motor-speed sensor,

motor-temperature sensor, current/power sensing, maximum/minimum motor power delivery settings, maximum/minimum motor acceleration and/or deceleration, or any of a variety of performance parameters used in industrial motor controls. The electrical power delivered to the motor **2978** via the motor controller **2976** generates motion transferred to the pump **2980** via a transmission **2982**. The pump **2980** moves hydraulic fluid into the hydraulic inlet port **2940** and out of the hydraulic outlet port **2942**. In one illustrative configuration the pump **2980** is AC 5 kW motor operating at about 600 amp.

(99) With reference to FIG. **35** showing top plan view of the control system **2900**, the motor **2978** is mechanically coupled to the pump **2980** via the transmission **2982**. In one illustrative configuration, the transmission **2982** may include an inlet sprocket **2984**, an input flange bearing **2986**, a chain **2988**, an output sprocket **2990**, an output flange bearing **2992**. The inlet sprocket **2984** and the input flange bearing **2986** cooperate to transfer mechanical power to the chain **2988**. The output sprocket **2990** and the output flange bearing **2992** cooperate to transfer mechanical power from the chain **2988** to the pump **2980**.

(100) With reference to FIG. **36** showing a side elevation view of the control system **2900** with the (illustrative) enclosure removed, the control system **2900** may be provided with a first fan **2994** adjacent to the fan shroud **2932** (FIG. **35**) and with a second fan **2996** adjacent to the fan shroud **2936** (FIG. **35**). As previously described, the cooling flow path **2938** (FIG. **31**) may be augmented by one or more fans such as fan **2994** and second fan **2996**.

(101) With reference to FIG. **37** showing perspective view of the enclosure **2902** of the control system **2900** with most components removed, the enclosure **2902** may include battery partitions **3700**, a first fan opening **3702**, a second fan opening **3704**, a bearing flange backer plate **3706**, and a plurality of mounting studs **3708**. The first fan opening **3702** may receive and provide fluid communication through the first fan **2994**. The second fan opening **3704** may receive and provide fluid communication through the second fan **2996**. The bearing flange backer plate **3706** may be captured by attachment hardware for mounting the input flange bearing **2986** (FIG. **35**) and the output sprocket **2990** (FIG. **35**) to the front **2918**. The mounting studs **3708** are formed (e.g., spot welded) on the front **2918** for receiving the circuit enclosure **2930** (as best illustrated in FIG. **30**).

(102) With reference to FIG. **38** showing perspective view of the circuit enclosure **2930**, the circuit enclosure **2930** may house the circuit board assembly **2960** (FIG. **33**) as previously described and/or may further house additional hydraulic components such as one or more solenoids. In some configurations, the solenoids for operating the hydraulic, left hydraulic cylinder **700**, and the right hydraulic cylinder **750** may be positioned inside the circuit enclosure **2930**. If configured with solenoids inside the circuit enclosure **2930**, a mounting bracket **3800** may be formed in/on the circuit enclosure **2930** for attaching solenoids thereto.

(103) When utilizing the control system **2900** as part of the dump trailer **100**, the control system **2900** may receive power intermittently or continuously. If power is provided continuously, the tractor unit **12** may have a robust alternator capable of supplying operating power (for the internal combustion engine) as well as to charge the battery **2972** (FIG. **32**). Alternatively, continuous power supply may be from an appropriately-sized (i.e., relatively small) generator operating on fossil fuel or solar power. If this appropriately-sized generator is supplied, it would be located (permanently or temporarily) somewhere on the trailer itself. When the user invokes a process of controlling or dumping material from the bed of the trailer, the power from the battery(ies) is transferred to the motor. In response, hydraulic fluid is pressurized by the pump.

(104) With reference to FIG. **39** showing a side elevation view of a side-dump trailer **3900**, the side-dump trailer **3900** may be provided with one or more tubs (such as the illustrated configuration with 2 separately operating tubs). When configured with the side-dump trailer **3900**, a tub **3902** may take any shape but is pivotally attached to a frame **3904** along a longitudinal axis **3906**. As shown in FIG. **40**, showing a perspective view of the front of the side-dump trailer **3900**, the side-dump trailer **3900** may be fitted with a front hydraulic cylinder **3910**. The front hydraulic

cylinder **3910** may have a proximal end **3912** and an oppositely disposed distal end **3914**. The proximal end **3912** is attached to the frame **3904** while the oppositely disposed distal end **3914** is attached to the tub **3902**. With reference to FIG. **41**, showing a perspective view of from the rear of the side-dump trailer **3900**, a rear tub **3908** is similarly attached to the frame **3904** by the longitudinal axis **3906** (FIG. **36**). The rear tub **3908** may be fitted with a rear hydraulic cylinder **3920**. The rear hydraulic cylinder **3920** may have a proximal end **3922** and an oppositely disclosed distal end **3924**. The proximal end **3922** is attached to the frame **3904** while the oppositely disposed distal end **3924** is attached to the tub **3908**. Due to limitations of the illustration(s) of FIGS. **39-41**, it is to be understood that hydraulic cylinders on each end of a provided tub may retain the orientation, attachment, and operation as the front hydraulic cylinder **3910** and the rear hydraulic cylinder **3920**. Therefore, if the rear tub **3908** is moved to unload material contained therein, the front hydraulic cylinder **3910** and the rear hydraulic cylinder **3920** receive hydraulic fluid to expand in length (therefore proximal end **3912** moves away from oppositely disposed distal end **3914**; and, proximal end **3922** moves away from the oppositely disclosed distal end **3924**). This expansion of the front hydraulic cylinder **3910** and the rear hydraulic cylinder **3920** causes the tub **3908** and/or tub **3902** to rotate about the longitudinal axis **3906**. If provided with the highly accurate control system **2900** (FIG. **30**), the length and speed of expansion of the hydraulic cylinders **3910**, **3920** is controlled and the tub **3908** (for example) is generally not exposed to twisting forces.

(105) In an alternative configuration illustrated, for example in FIG. **33**, the telemetry device **2966** of the circuit enclosure **2930** may receive instructions from a controller of the trailer (e.g., owner, regulatory/governing body, on-site personnel manager, safety manager, etc.) to block/stop/intermittently-interrupt usage of the trailer. In this illustrative configuration, a computer device may forward information to one or more components of the control system **2900**—for example, the shunt **2974** may be configured as a switch that is pulled open upon receiving instructions from the server. Alternatively, the motor controller **2976** (FIG. **34**) may limit the speed or outright prohibit use of the trailer until the controller issues new instructions (or other termination event such as a time-based block on usage). When the trailer is locked by the controller, certain features may remain operable for safety reasons (e.g., lowering the trailer to enable movement of the tractor body, relocation to a different site, operation of standard or parking breaks, etc.).

(106) In an alternative configuration illustrated in FIG. **35**, the motor controller **2976** may be configured with an over-current system capable of sensing, reporting, and/or acting on potentially damaging situations. For example, if the trailer is being raised and an overhead object is blocking movement of the trailer, the pressure in the hydraulics will increase. If other control features are not present, the pressure ‘call’ on the pump and motor is high. The requested/situational increase in pressure likely causes an increase in power from the batteries to run the motor. Because the voltage is typically maintained, this requires an increase in current. The increase in current can be sensed and may result in termination (or reversal) of the event causing the spike in power.

(107) In an alternative configuration illustrated in FIGS. **33** and **34**, the control system **2900** may be configured with a powered-only displacement feature whereby the motor **2978** must be actively moved forward and/or reverse to move the left hydraulic cylinder **700** and/or the right hydraulic cylinder **750**. This optional feature may provide security from unintended movement of the trailer in the event of a mechanical or electrical failure. For example, if the battery **2972** is fully depleted, the motor **2978** (and the pump **2980**) will not allow hydraulic fluid to move into or out of the cylinders **700** and **750**.

(108) With reference to FIG. **42** showing a side elevation view of the semi-trailer truck **10**, operating between the transporting condition (similar to FIG. **2**), and a dumping condition (similar to FIG. **3**), as an alternative configuration. As illustrated earlier, the draft arm assembly **600** may be rotatably attached to the tub **200**. Alternatively, the draft arm assembly **600** may also be connected

to the rear assembly **300**. As may be appreciated, the draft arm assembly **600** may be connected to the pyramid **430** of the rear assembly **300**. In this configuration, the proximal end of the draft arm assembly **600** may be connected to the coupling plate **500** in a similar manner as illustrated by FIG. 26.

(109) With continued reference to FIG. 42, the connection of the proximal end of the draft arm assembly **600** with the coupling plate **500**, and the distal end of the draft arm assembly **600** to the pyramid **430**, may result in constraining the draft arm assembly **600** between the pyramid **430**, and the coupling plate **500**, i.e., fixating the draft arm assembly **600** between the pyramid **430** and the coupling plate **500**. In such configuration, the draft arm assembly **600** may be fixated in such a manner that the draft arm assembly **600** may be parallel to the ground **20**. Therefore, the parallel arrangement of the draft arm assembly **600** may result in the draft arm angle **604** at 0° .

(110) Therefore, with reference to FIG. 42, the draft arm assembly **600** may be stationary and parallel to the ground even when the semi-trailer truck **10** operates from the transporting condition to the dumping condition. For example, during transporting condition, the tub **200** may be at stationary condition, and rested on the draft arm assembly **600**. And during dumping condition, the left cylinder length and the right cylinder length is increased to extend the left hydraulic cylinder **700** and right hydraulic cylinder **750** respectively, thereby rotating the tub **200** about the rear pivot.

(111) In another alternative configuration, to prevent rollover event of the semi-trailer truck **10** when the draft arm assembly **600** is stationary, an orientation sensor, or a position sensor (not shown in figures) may be assembled on the draft arm assembly **600**, and connected to the control system **800**. The orientation sensor may be configured to indicate, or sense in real time, orientation of the draft arm assembly **600**. Further, any change in the orientation of the draft arm assembly **600** in a rollover condition, may be sensed by the orientation sensor, and therefore, may generate a signal to the control system **800** initiating onset of a rollover event. For example, a current orientation in x,y,z coordinates of the draft arm assembly **600** may be determined, and compared against a threshold for each of the x, y, and z coordinate. Therefore, the control system **800** may operate the left hydraulic cylinder **700**, or the right hydraulic cylinder **750**, to prevent the rollover event.

(112) While the above description includes terms such as top, bottom, left, right, inside, outside, front, back, and other descriptors regarding physical orientation and/or position, it is to be understood that these are provided for illustrative purposes only. However, the present description was provided to convey to one skilled in the art.

(113) As used herein, the terms symmetrical, symmetrically, symmetrically disposed and variants similar to these is to be construed as generally a mirror copy about a central plane.

(114) While the principles of the disclosure have been described above in connection with the specific apparatuses and methods, it is to be understood that this description is made only by way of example and not as limitation on the scope of the disclosure.

Various Notes and Examples

(115) Example 1 is a dump trailer for a semi-trailer truck comprising: a tub, defining an exterior portion, comprising: a web; a left wall adjoining the web; a right wall adjoining the web and symmetrical to the left wall; a closed end adjoining the web, left wall, and right wall; an open end oppositely disposed from the closed end and adjoining the web, left wall, and right wall; a hinge axis parallel to and offset from the web; a left hinge formed on the exterior portion of the left wall and coaxial with the hinge axis; a right hinge formed on the exterior portion of the right wall and coaxial with the hinge axis; a distal draft arm axis parallel and adjacent to the web, the distal draft arm axis formed on the exterior portion of the web; and a center hinge formed on the web and coaxial with the distal draft arm axis, the center hinge symmetrically disposed between the left wall and the right wall; a coupling plate to engage the dump trailer to the semi-trailer truck, comprising: a hydraulic cylinder axis formed on the coupling plate; a left hydraulic cylinder hinge formed on the coupling plate and coaxial with the hydraulic cylinder axis; a right hydraulic cylinder hinge

formed on the coupling plate and coaxial with the hydraulic cylinder axis; a proximal draft arm axis formed on the coupling plate; and a draft arm hinge formed on the coupling plate and coaxial with the draft arm axis, the draft arm hinge symmetrically disposed between the left hydraulic cylinder hinge and the left hydraulic cylinder hinge; a draft arm symmetrically disposed between the left wall and the right wall of the tub, the draft arm comprising: a proximal end attached to the draft arm hinge of the coupling plate and rotationally attached to the coupling plate about the proximal draft arm axis; and a distal end oppositely disposed from the proximal end, attached to the center hinge of the tub, and rotationally attached to the tub about the distal draft arm axis; a left hydraulic cylinder defining a left cylinder length, the left hydraulic cylinder comprising: a proximal end attached to the left hydraulic cylinder hinge; a distal end oppositely disposed from the proximal end and attached to the left hinge on the tub; and wherein addition of a first amount of fluid changes the left cylinder length; a right hydraulic cylinder defining a right cylinder length, the right hydraulic cylinder comprising: a proximal end attached to the right hydraulic cylinder hinge; a distal end oppositely disposed from the proximal end and attached to the right hinge on the tub; and wherein addition of a second amount of fluid changes the right cylinder length; a rear assembly pivotally attached to the tub at the web adjacent to the open end, the rear assembly comprising: a first pair of wheels; and a rear pivot around which the tub rotates; whereby increase in both left cylinder length and right cylinder length cause: the distal end of the draft arm to move away from earth; and the tub to rotate about the rear pivot.

(116) In Example 2, the subject matter of Example 1 optionally includes and further comprising: a coupling plane that is parallel to the coupling plate; a draft arm plane that is coplanar to the draft arm, wherein the proximal draft arm axis and distal draft arm axis reside in the draft arm plane; and wherein the coupling plane and the draft arm plane are non-parallel.

(117) In Example 3, the subject matter of any one or more of Examples 1-2 optionally include wherein: the left hydraulic cylinder comprises: a left bore cylinder at the proximal end, the left bore cylinder attached to the left hydraulic cylinder hinge; a left rod slidingly coupled to the left bore cylinder, the left rod attached to the left hinge on the tub; and wherein addition of the first amount of fluid ejects the left rod from the left bore cylinder; the right hydraulic cylinder comprises: a right bore cylinder at the proximal end, the right bore cylinder attached to the right hydraulic cylinder hinge; a right rod slidingly coupled to the right bore cylinder, the right rod attached to the right hinge on the tub; and wherein addition of the second amount of fluid ejects the right rod from the right bore cylinder.

(118) In Example 4, the subject matter of any one or more of Examples 1-3 optionally include wherein the hydraulic cylinder axis and the proximal draft arm axis are coaxial.

(119) In Example 5, the subject matter of any one or more of Examples 1-4 optionally include wherein the hinge axis is formed through the left wall and the right wall.

(120) In Example 6, the subject matter of Example 5 optionally includes and further comprising: a topside formed on the tub; wherein the left hydraulic cylinder hinge is formed on the left wall adjoining the topside of the tub; and wherein the right hydraulic cylinder hinge is formed on the right wall adjoining the topside of the tub.

(121) In Example 7, the subject matter of any one or more of Examples 1-6 optionally include and further comprising: a hydraulic pump attached to the dump trailer; an accumulator in fluid communication with the hydraulic pump and attached to the dump trailer; a left hydraulic valve in fluid communication with the left hydraulic cylinder; a right hydraulic valve in fluid communication with the right hydraulic cylinder; a controller in communication with the left hydraulic valve and the right hydraulic valve; an electric power supply line in electrical communication with semi-trailer truck and the hydraulic pump; whereby power from semi-trailer truck is transferred to the hydraulic pump for pressurizing an amount of hydraulic fluid; whereby at least a portion of the hydraulic fluid is transferred to the right hydraulic cylinder via the right hydraulic valve and the accumulator; and whereby at least a portion of the hydraulic fluid is

transferred to the left hydraulic cylinder via the left hydraulic valve and the accumulator.

(122) In Example 8, the subject matter of any one or more of Examples 1-7 optionally include and further comprising: a center plane formed on the tub, wherein the center plane is symmetrically located between the left wall and the right wall; a chassis plane formed on semi-trailer truck, wherein the chassis plane is defined by at least two axles of the semi-trailer truck that are coplanar to the chassis plane; a perpendicular condition wherein: the center plane of the tub is perpendicular to the chassis plane; and the left cylinder length and the right cylinder length are equal; a canted condition wherein: the center plane is non-perpendicular to the chassis plane; and the right cylinder length and the right cylinder length are different.

(123) In Example 9, the subject matter of any one or more of Examples 1-8 optionally include wherein the rear assembly further comprises: a second pair of wheels parallel to the first pair of wheels; a spring assembly comprising: at least one leaf spring comprising first end and a second end; a first axle attachment on the first end; a second axle attachment on the second end; and a hinge symmetrically attached between the first end and the second end of the leaf spring; wherein the first pair of wheels are attached to the first axle attachment with a first axle; wherein the second pair of wheels are attached to the second axle attachment with a second axle; and wherein the rear pivot about which the tub rotates is coaxial with a hinge assembly.

(124) Example 10 is a method of controlling unloading of a dump trailer comprising: providing a tub, defining an exterior portion, comprising: a web; a left wall adjoining the web; a right wall adjoining the web and symmetrical to the left wall; a closed end adjoining the web, left wall, and right wall; an open end: oppositely disposed from the closed end; and, adjoining the web, left wall, and right wall; a hinge axis formed through the left wall and the right wall; a left hinge formed on the exterior portion of the left wall and coaxial with the hinge axis; a right hinge formed on the exterior portion of the right wall and coaxial with the hinge axis; a distal draft arm axis adjacent to the web and on the exterior portion of the web; a center hinge formed on the web and coaxial with the distal draft arm axis; and a center plane symmetrically disposed between the left wall and the right wall; providing a coupling plate, configured to engage the dump trailer to semi-trailer truck, comprising: a hydraulic cylinder axis formed on the coupling plate; a left hydraulic cylinder hinge: formed on the coupling plate; and coaxial with the hydraulic cylinder axis; a right hydraulic cylinder hinge: formed on the coupling plate; and coaxial with the hydraulic cylinder axis; a proximal draft arm axis formed on the coupling plate; and a draft arm hinge: formed on the coupling plate; and coaxial with the draft arm axis; providing a draft arm, symmetrically disposed between the left wall and the right wall of the tub, comprising: a proximal end rotationally attached to the coupling plate about the proximal draft arm axis; and a distal end: oppositely disposed from the proximal end; attached to the center hinge; and whereby the draft arm is rotationally attached to the tub about the distal draft arm axis; providing a left hydraulic cylinder, defining a left cylinder length, comprising: a proximal end attached to the left hydraulic cylinder hinge; a distal end oppositely disposed from the proximal end and attached to the left hinge on the tub; and wherein addition of a first amount of fluid changes the left cylinder length; providing a right hydraulic cylinder, defining a right cylinder length, comprising: a proximal end attached to the right hydraulic cylinder hinge; a distal end oppositely disposed from the proximal end and attached to the right hinge on the tub; and wherein addition of a second amount of fluid changes the right cylinder length; providing a rear assembly attached to the tub adjoining the web and adjacent to the open end, the rear assembly comprising: a first pair of wheels; and a rear pivot about which the tub rotates; a second pair of wheels parallel to the first pair of wheels; a spring assembly comprising: at least one leaf spring comprising first end and a second end; a first axle attachment on the first end; a second axle attachment on the second end; and a hinge symmetrically attached between the first end and the second end of the leaf spring; wherein the first pair of wheels are attached to the first axle attachment with a first axle; wherein the second pair of wheel are attached to the second axle attachment with a second axle; and wherein the rear pivot about which the tub rotates is coaxial

with the hinge of the rear assembly; increasing in both the left cylinder length and the right cylinder length: moves the distal end of the draft arm to away from earth; and rotates the tub about the rear pivot; monitoring perpendicularity of the center plane; increasing either the left cylinder length or the right cylinder length to compress the leaf spring in the spring assembly; and thereby controlling the unloading of the dump trailer.

(125) In Example 11, the subject matter of Example 10 optionally includes and further comprising, providing: a hydraulic pump attached to the dump trailer; an accumulator in fluid communication with the hydraulic pump and attached to the dump trailer; a left hydraulic valve in fluid communication with the left hydraulic cylinder; a right hydraulic valve in fluid communication with the right hydraulic cylinder; a controller in communication with the left hydraulic valve and the right hydraulic valve; an electric power supply line in electrical communication with semi-trailer truck and the hydraulic pump; transferring power from semi-trailer truck to the hydraulic pump; pressurizing an amount of hydraulic fluid with the hydraulic pump; whereby during the monitoring perpendicularity, the controller independently controls the left hydraulic valve and the right hydraulic valve; and thereby controlling unloading of the dump trailer.

(126) Example 12 is a dump trailer for a semi-trailer truck comprising: a chassis plane formed on semi-trailer truck, wherein the chassis plane is defined by at least two axles of the semi-trailer truck that are coplanar to the chassis plane; a tub, comprising: a web; a left side adjoining the web; a right side adjoining the web; and a center plane formed on the tub, wherein the center plane is symmetrically located between the left wall and the right wall; a draft arm symmetrically disposed between the left side and the right side of the tub, the draft arm comprising: a proximal end; and a distal end oppositely disposed from the proximal end, rotationally attached to the web of the tub; a coupling plate to engage the dump trailer to the semi-trailer truck, wherein: the proximal end of the draft arm is rotationally attached to the coupling plate; a left hydraulic cylinder defining a left cylinder length, the left hydraulic cylinder comprising: a proximal end rotationally attached to the coupling plate; and a distal end oppositely disposed from the proximal end and rotationally attached to the tub; a right hydraulic cylinder defining a right cylinder length, the right hydraulic cylinder comprising: a proximal end rotationally attached to the coupling plate; and a distal end oppositely disposed from the proximal end and rotationally attached to the tub; a rear assembly pivotally attached to the tub at the web, the rear assembly comprising: a first pair of wheels; and a rear pivot around which the tub rotates; a perpendicular condition wherein: the center plane of the tub is perpendicular to the chassis plane; and the left cylinder length and the right cylinder length are equal; a canted condition wherein: the center plane is non-perpendicular to the chassis plane; and the right cylinder length and the right cylinder length are different.

(127) In Example 13, the subject matter of Example 12 optionally includes wherein the draft arm and the coupling plane are non-parallel.

(128) In Example 14, the subject matter of any one or more of Examples 12-13 optionally include wherein: the left hydraulic cylinder comprises: a left bore cylinder rotationally attached to coupling plate; and a left rod slidingly coupled to the left bore cylinder, the left rod rotationally attached to the left side of the tub; the right hydraulic cylinder comprises: a right bore cylinder rotationally attached to the coupling plate; and a right rod slidingly coupled to the right bore cylinder, the right rod rotationally attached to the right side on the tub.

(129) In Example 15, the subject matter of Example 14 optionally includes and further comprising: a topside formed on the tub; wherein the left rod of the left hydraulic cylinder is adjoining the topside of the tub; and wherein the right rod of the right hydraulic cylinder is adjoining the topside of the tub.

(130) In Example 16, the subject matter of any one or more of Examples 12-15 optionally include wherein the left hydraulic cylinder, the right hydraulic cylinder, and the proximal end of the draft arm are rotationally attached to the coupling plate along a common axis of rotation.

(131) In Example 17, the subject matter of any one or more of Examples 12-16 optionally include

and further comprising: a hydraulic pump attached to the dump trailer; an accumulator in fluid communication with the hydraulic pump and attached to the dump trailer; a left hydraulic valve in fluid communication with the left hydraulic cylinder; a right hydraulic valve in fluid communication with the right hydraulic cylinder; a controller in communication with the left hydraulic valve and the right hydraulic valve; an electric power supply line in electrical communication with semi-trailer truck and the hydraulic pump; whereby power from semi-trailer truck is transferred to the hydraulic pump for pressurizing an amount of hydraulic fluid; whereby at least a portion of the hydraulic fluid is transferred to the right hydraulic cylinder via the right hydraulic valve and the accumulator; and whereby at least a portion of the hydraulic fluid is transferred to the left hydraulic cylinder via the left hydraulic valve and the accumulator.

(132) In Example 18, the subject matter of Example 17 optionally includes and further comprising: an accelerometer attached to the dump trailer configured to sense orientation of the center plane; wherein the accelerometer is in communication with the controller; and a set of instructions implemented by the controller to change the right cylinder length and the left cylinder length.

(133) In Example 19, the subject matter of Example 18 optionally includes wherein the accelerometer is attached to the web of the tub.

Claims

1. A dump trailer for a semi-trailer truck comprising: a tub, defining an exterior portion, comprising: a web; a left wall adjoining the web; a right wall adjoining the web and symmetrical to the left wall; a closed end adjoining the web, the left wall, and the right wall; an open end oppositely disposed from the closed end and adjoining the web, the left wall, and the right wall; a hinge axis parallel to and offset from the web; a left hinge formed on the exterior portion of the left wall and coaxial with the hinge axis; a right hinge formed on the exterior portion of the right wall and coaxial with the hinge axis; a distal draft arm axis parallel and adjacent to the web, the distal draft arm axis formed on the exterior portion of the web; and a center hinge formed on the web and coaxial with the distal draft arm axis, the center hinge symmetrically disposed between the left wall and the right wall; a coupling plate to engage the dump trailer to the semi-trailer truck, comprising: a hydraulic cylinder axis formed on the coupling plate; a left hydraulic cylinder hinge formed on the coupling plate and coaxial with the hydraulic cylinder axis; a right hydraulic cylinder hinge formed on the coupling plate and coaxial with the hydraulic cylinder axis; a proximal draft arm axis formed on the coupling plate; and a draft arm hinge formed on the coupling plate and coaxial with the proximal draft arm axis and the distal draft arm axis, the draft arm hinge symmetrically disposed between the left hydraulic cylinder hinge and the right hydraulic cylinder hinge; a left hydraulic cylinder defining a left cylinder length, the left hydraulic cylinder comprising: a proximal end attached to the left hydraulic cylinder hinge; and a distal end oppositely disposed from the proximal end and attached to the left hinge on the tub; wherein addition of a first amount of fluid changes the left cylinder length; a right hydraulic cylinder defining a right cylinder length, the right hydraulic cylinder comprising: a proximal end attached to the right hydraulic cylinder hinge; and a distal end oppositely disposed from the proximal end and attached to the right hinge on the tub; wherein addition of a second amount of fluid changes the right cylinder length; a rear assembly pivotally attached to the tub at the web adjacent to the open end, the rear assembly comprising: a first pair of wheels; and a rear pivot around which the tub rotates; wherein the dump trailer is configured to operate between: a transporting condition; wherein the tub is in a stationary position; a dumping condition; wherein the left cylinder length and the right cylinder length is increased to extend the left hydraulic cylinder and the right hydraulic cylinder respectively; whereby increase in both the left cylinder length and the right cylinder length cause: the tub to rotate about the rear pivot.

2. The dump trailer of claim 1, further comprising: a draft arm symmetrically disposed between the

left wall and the right wall of the tub, the draft arm comprising: a proximal end attached to the draft arm hinge of the coupling plate and attached to the coupling plate about the proximal draft arm axis; and a distal end oppositely disposed from the proximal end.

3. The dump trailer of claim 2, wherein the distal end of the draft arm is configured to adjoin the rear assembly, and the draft arm is in a stationary position parallel to ground when the tub is rotated about the rear pivot.

4. The dump trailer of claim 2, and further comprising: a coupling plane that is parallel to the coupling plate; and a draft arm plane that is coplanar to the draft arm, wherein the proximal draft arm axis and the distal draft arm axis reside in the draft arm plane; wherein the coupling plane and the draft arm plane are parallel.

5. The dump trailer of claim 1, wherein: the left hydraulic cylinder comprises: a left bore cylinder at the proximal end, the left bore cylinder attached to the left hydraulic cylinder hinge; and a left rod slidingly coupled to the left bore cylinder, the left rod attached to the left hinge on the tub; wherein addition of the first amount of fluid ejects the left rod from the left bore cylinder; and the right hydraulic cylinder comprises: a right bore cylinder at the proximal end, the right bore cylinder attached to the right hydraulic cylinder hinge; and a right rod slidingly coupled to the right bore cylinder, the right rod attached to the right hinge on the tub; wherein addition of the second amount of fluid ejects the right rod from the right bore cylinder.

6. The dump trailer of claim 1, wherein the hydraulic cylinder axis and the proximal draft arm axis are coaxial.

7. The dump trailer of claim 1, wherein the hinge axis is formed through the left wall and the right wall.

8. The dump trailer of claim 1, and further comprising: a topside formed on the tub; and wherein the left hydraulic cylinder hinge is formed on the left wall adjoining the topside of the tub; wherein the right hydraulic cylinder hinge is formed on the right wall adjoining the topside of the tub.

9. The dump trailer of claim 1 and further comprising: a hydraulic pump attached to the dump trailer; an accumulator in fluid communication with the hydraulic pump and attached to the dump trailer; a left hydraulic valve in fluid communication with the left hydraulic cylinder; a right hydraulic valve in fluid communication with the right hydraulic cylinder; a controller in communication with the left hydraulic valve and the right hydraulic valve; an electric power supply line in electrical communication with the semi-trailer truck and the hydraulic pump; whereby power from the semi-trailer truck is transferred to the hydraulic pump operable to pressurize an amount of hydraulic fluid; whereby at least a portion of the fluid is transferred to the right hydraulic cylinder via the right hydraulic valve and the accumulator; and whereby at least a portion of the fluid is transferred to the left hydraulic cylinder via the left hydraulic valve and the accumulator.

10. The dump trailer of claim 1 and further comprising: a center plane formed on the tub, wherein the center plane is symmetrically located between the left wall and the right wall; a chassis plane formed on the semi-trailer truck, wherein the chassis plane is defined by at least two axles of the semi-trailer truck that are coplanar to the chassis plane; a perpendicular condition wherein: the center plane of the tub is perpendicular to the chassis plane; and the left cylinder length and the right cylinder length are equal; and a canted condition wherein: the center plane is non-perpendicular to the chassis plane; and the right cylinder length and the right cylinder length are different.

11. The dump trailer of claim 1 wherein the rear assembly further comprises: a second pair of wheels parallel to the first pair of wheels; and a spring assembly comprising: at least one leaf spring comprising first end and a second end; a first axle attachment on the first end; a second axle attachment on the second end; a hinge symmetrically attached between the first end and the second end of the at least one leaf spring; wherein the first pair of wheels are attached to the first axle attachment with a first axle; and wherein the second pair of wheels are attached to the second axle

attachment with a second axle; wherein the rear pivot about which the tub rotates is coaxial with a hinge assembly.

12. A method of controlling unloading of a dump trailer comprising: providing a tub, defining an exterior portion, comprising: a web; a left wall adjoining the web; a right wall adjoining the web and symmetrical to the left wall; a closed end adjoining the web, left wall, and right wall; an open end: oppositely disposed from the closed end; and, adjoining the web, the left wall, and the right wall; a hinge axis formed through the left wall and the right wall; a left hinge formed on the exterior portion of the left wall and coaxial with the hinge axis; a right hinge formed on the exterior portion of the right wall and coaxial with the hinge axis; a distal draft arm axis adjacent to the web and on the exterior portion of the web; a center hinge formed on the web and coaxial with the distal draft arm axis; and a center plane symmetrically disposed between the left wall and the right wall; providing a coupling plate, configured to engage the dump trailer to a semi-trailer truck, comprising: a hydraulic cylinder axis formed on the coupling plate; a left hydraulic cylinder hinge: formed on the coupling plate; and coaxial with the hydraulic cylinder axis; a right hydraulic cylinder hinge: formed on the coupling plate; and coaxial with the hydraulic cylinder axis; a proximal draft arm axis formed on the coupling plate; and a draft arm hinge formed on the coupling plate; and coaxial with the proximal draft arm axis and the distal draft arm axis; providing a left hydraulic cylinder, defining a left cylinder length, comprising: a proximal end attached to the left hydraulic cylinder hinge; and a distal end oppositely disposed from the proximal end and attached to the left hinge on the tub; wherein addition of a first amount of fluid changes the left cylinder length; providing a right hydraulic cylinder, defining a right cylinder length, comprising: a proximal end attached to the right hydraulic cylinder hinge; and a distal end oppositely disposed from the proximal end and attached to the right hinge on the tub; wherein addition of a second amount of fluid changes the right cylinder length; providing a rear assembly attached to the tub adjoining the web and adjacent to the open end, the rear assembly comprising: a first pair of wheels; a rear pivot about which the tub rotates; and a second pair of wheels parallel to the first pair of wheels; a spring assembly comprising: at least one leaf spring comprising first end and a second end; a first axle attachment on the first end; and a second axle attachment on the second end; a hinge symmetrically attached between the first end and the second end of the at least one leaf spring; wherein the first pair of wheels are attached to the first axle attachment with a first axle; and wherein the second pair of wheel are attached to the second axle attachment with a second axle; wherein the rear pivot about which the tub rotates is coaxial with the hinge of the rear assembly; operating the dump trailer between a transporting condition and a dumping condition, wherein the transporting condition, the tub is in a stationary position; and in the dumping condition, increasing the left cylinder length and the right cylinder length for extending the left hydraulic cylinder and the right hydraulic cylinder respectively; and increasing in both the left cylinder length and the right cylinder length: rotating the tub about the rear pivot; compressing the at least one leaf spring in the spring assembly thereby controlling the unloading of the dump trailer.

13. The method of controlling unloading of the dump trailer of claim 12, further comprising: providing a draft arm, symmetrically disposed between the left wall and the right wall of the tub, comprising: a proximal end attached to the draft arm hinge of the coupling plate and rotationally attached to the coupling plate about the proximal draft arm axis; and a distal end oppositely disposed from the proximal end.

14. The method of controlling unloading of the dump trailer of claim 13, further comprising: adjoining the draft arm to the rear assembly; and positioning the draft arm parallel to ground when the tub is rotated about the rear pivot.

15. The method of controlling unloading of the dump trailer of claim 12 and further comprising: providing: a hydraulic pump attached to the dump trailer; an accumulator in fluid communication with the hydraulic pump and attached to the dump trailer; a left hydraulic valve in fluid communication with the left hydraulic cylinder; a right hydraulic valve in fluid communication

with the right hydraulic cylinder; a controller in communication with the left hydraulic valve and the right hydraulic valve; and an electric power supply line in electrical communication with the semi-trailer truck and the hydraulic pump; transferring power from the semi-trailer truck to the hydraulic pump; pressurizing an amount of hydraulic fluid with the hydraulic pump; monitoring perpendicularity of the center plane; whereby during monitoring perpendicularity of the center plane, the controller independently controls the left hydraulic valve and the right hydraulic valve; and thereby controlling unloading of the dump trailer.

16. The method of controlling unloading of the dump trailer of claim 13, and further comprising: providing: a coupling plane parallel to the coupling plate; and a draft arm plane, coplanar to the draft arm, wherein the proximal draft arm axis and the distal draft arm axis reside in the draft arm plane; wherein the coupling plane and the draft arm plane are parallel.

17. The method of controlling unloading of the dump trailer of claim 12, wherein: providing the left hydraulic cylinder comprises: a left bore cylinder at the proximal end, the left bore cylinder attached to the left hydraulic cylinder hinge; and a left rod slidingly coupled to the left bore cylinder, the left rod attached to the left hinge on the tub; wherein addition of the first amount of fluid ejects the left rod from the left bore cylinder; and providing the right hydraulic cylinder comprises: a right bore cylinder at the proximal end, the right bore cylinder attached to the right hydraulic cylinder hinge; and a right rod slidingly coupled to the right bore cylinder, the right rod attached to the right hinge on the tub; wherein addition of the second amount of fluid ejects the right rod from the right bore cylinder.

18. The method of controlling unloading of the dump trailer of claim 12 and further comprising: providing: a topside on the tub, wherein the topside is formed on the tub; and wherein the left hydraulic cylinder hinge is formed on the left wall adjoining the topside of the tub; wherein the right hydraulic cylinder hinge is formed on the right wall adjoining the topside of the tub.

19. The method of controlling unloading of the dump trailer of claim 12, wherein the hydraulic cylinder axis and the proximal draft arm axis are coaxial.

20. The method of controlling unloading of the dump trailer of claim 12, wherein the hinge axis is formed through the left wall and the right wall.

21. The method of controlling unloading of the dump trailer of claim 12, further comprising: providing: a center plane on the tub, wherein the center plane is symmetrically formed on the tub and located between the left wall and the right wall; and a chassis plane on the semi-trailer truck, wherein the chassis plane is defined by at least two axles of the semi-trailer truck that are coplanar to the chassis plane; operating the dump trailer in: a perpendicular condition wherein: the center plane of the tub is perpendicular to the chassis plane; and the left cylinder length and the right cylinder length are equal; and a canted condition wherein: the center plane is non-perpendicular to the chassis plane; and the left cylinder length and the right cylinder length are different.

22. The method of controlling unloading of the dump trailer of claim 12, wherein: providing the rear assembly further comprising: a second pair of wheels parallel to the first pair of wheels; and a spring assembly comprising: at least one leaf spring comprising first end and a second end; a first axle attachment on the first end; a second axle attachment on the second end; a hinge symmetrically attached between the first end and the second end of the at least one leaf spring; wherein the first pair of wheels are attached to the first axle attachment with a first axle; and wherein the second pair of wheels are attached to the second axle attachment with a second axle; wherein the rear pivot about which the tub rotates is coaxial with a hinge assembly.

23. A dump trailer for a semi-trailer truck comprising: a chassis plane formed on the semi-trailer truck, wherein the chassis plane is defined by at least two axles of the semi-trailer truck that are coplanar to the chassis plane; a tub, comprising: a web; a left side adjoining the web; a right side adjoining the web; and a center plane formed on the tub, wherein the center plane is symmetrically located between the left side and the right side; a draft arm symmetrically disposed between the left side and the right side of the tub, the draft arm comprising: a proximal end; and a distal end

oppositely disposed from the proximal end; a coupling plate to engage the dump trailer to the semi-trailer truck, wherein: the proximal end of the draft arm is attached to the coupling plate; a left hydraulic cylinder defining a left cylinder length, the left hydraulic cylinder comprising: a proximal end rotationally attached to the coupling plate; and a distal end oppositely disposed from the proximal end and rotationally attached to the tub; a right hydraulic cylinder defining a right cylinder length, the right hydraulic cylinder comprising: a proximal end rotationally attached to the coupling plate; and a distal end oppositely disposed from the proximal end and rotationally attached to the tub; a rear assembly pivotally attached to the tub at the web, the rear assembly comprising: a first pair of wheels; and a rear pivot around which the tub rotates; a perpendicular condition wherein: the center plane of the tub is perpendicular to the chassis plane; and the left cylinder length and the right cylinder length are equal; and a canted condition wherein: the center plane is non-perpendicular to the chassis plane; and the right cylinder length and the right cylinder length are different.

24. The dump trailer of claim 23, wherein the distal end of the draft arm is configured to adjoin the rear assembly, and the draft arm is in a stationary position parallel to ground when the tub is rotated about the rear pivot.

25. The dump trailer of claim 23 wherein the draft arm and the coupling plate are non-parallel.

26. The dump trailer of claim 23 wherein: the left hydraulic cylinder comprises: a left bore cylinder rotationally attached to coupling plate; and a left rod slidingly coupled to the left bore cylinder, the left rod rotationally attached to the left side of the tub; and the right hydraulic cylinder comprises: a right bore cylinder rotationally attached to the coupling plate; and a right rod slidingly coupled to the right bore cylinder, the right rod rotationally attached to the right side on the tub.

27. The dump trailer of claim 26 and further comprising: a topside formed on the tub; wherein the left rod of the left hydraulic cylinder is adjoining the topside of the tub; wherein the right rod of the right hydraulic cylinder is adjoining the topside of the tub.

28. The dump trailer of claim 23 wherein the left hydraulic cylinder, the right hydraulic cylinder; and the proximal end of the draft arm are rotationally attached to the coupling plate along a common axis of rotation.

29. The dump trailer of claim 23 and further comprising: a hydraulic pump attached to the dump trailer; an accumulator in fluid communication with the hydraulic pump and attached to the dump trailer; a left hydraulic valve in fluid communication with the left hydraulic cylinder; a right hydraulic valve in fluid communication with the right hydraulic cylinder; a controller in communication with the left hydraulic valve and the right hydraulic valve; an electric power supply line in electrical communication with the semi-trailer truck and the hydraulic pump; whereby power from the semi-trailer truck is transferred to the hydraulic pump operable to pressurize an amount of hydraulic fluid; whereby at least a portion of the hydraulic fluid is transferred to the right hydraulic cylinder via the right hydraulic valve and the accumulator; and whereby at least a portion of the hydraulic fluid is transferred to the left hydraulic cylinder via the left hydraulic valve and the accumulator.

30. The dump trailer of claim 29 and further comprising: an accelerometer attached to the dump trailer configured to sense orientation of the center plane; wherein the accelerometer is in communication with the controller; and a set of instructions implemented by the controller to change the right cylinder length and the left cylinder length.
